ProjFlow: Projection Sampling with Flow Matching for Zero-Shot Exact Spatial Motion Control

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Abstract

Precise spatial control in human motion generation commonly relies on task-specific retraining or slow inference-time optimization, and often fails to satisfy constraints exactly. We propose ProjFlow, a training-free projection sampler layered on a flow-matching motion prior that enforces hard spatial constraints in a zero-shot manner while preserving realism. At each time step, ProjFlow predicts a clean motion estimate and then projects it onto the feasible set of linear measurements using a kinematics-aware metric that distributes corrections coherently along the skeleton. A stochastic recomposition step mixes the corrected endpoint with noise to remain on the data manifold, and a time-varying pseudo-observation scheme with dynamic masking and adaptive variance robustly handles sparse inputs such as long-gap inpainting. On HumanML3D, ProjFlow achieves exact satisfaction (0.0000) of trajectory and location errors under control while maintaining competitive realism: for pelvis-trajectory control it reaches FID 0.107, improving over DNO (0.151) and matching trained controllers' realism without any retraining. When integrated with an ACMDM flow prior, ProjFlow attains 0.0000 MPJPE-2D in 2D-to-3D lifting with strong realism (FID 0.349/0.168 for Average/Cross settings), and consistently reduces residual errors left by trained baselines. Ablations show that removing the kinematics-aware metric, noise mixing, or pseudo-observations degrades FID to 1.152, 3.429, and 0.880 respectively, while maintaining 0.0000 control error, highlighting their importance for realism.

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1. Introduction

Controllable human motion generation has rapidly progressed with diffusion and flow-based models, yet precise spatial control—following user-specified joint trajectories, matching keyframes, or satisfying structure-aware constraints—remains challenging. Existing approaches typically rely on task-specific retraining (e.g., controller networks) or iterative inference-time optimization (e.g., noise optimization), which is slow, brittle, and often leaves nonzero residual errors. A zero-shot mechanism that enforces constraints exactly while preserving motion naturalness would enable interactive and reliable motion authoring in animation, VR/AR, and robotics.

This paper introduces ProjFlow, a projection-based sampler for flow-matching motion priors that provides exact spatial control without retraining or inner-loop optimization. The key insight is to cast many motion control tasks as linear inverse problems over global joint coordinates, enabling closed-form, data-consistent updates during sampling. Concretely, at each time step the sampler (i) predicts the clean motion endpoint via the flow prior, (ii) applies a closed-form projection that satisfies linear measurements using a kinematics-aware metric, and (iii) performs stochastic recomposition to remain on-manifold. A time-varying pseudo-observation schedule with dynamic masking and adaptive variance further stabilizes guidance from sparse inputs (e.g., long-gap inpainting).

ProjFlow delivers exact feasibility with strong realism across tasks. On HumanML3D[3], our projection sampler achieves 0.0000 trajectory/location errors under control while keeping low FID. In pelvis-trajectory control, ProjFlow attains FID 0.107, surpassing DNO[5] (0.151) and approaching trained controllers. In 2D-to-3D lifting conditioned on text, ProjFlow reaches 0.000 MPJPE-2D (exact reprojection) with FID 0.349/0.168 (Average/Cross). Ablations reveal the importance of the kinematics-aware metric, noise mixing, and pseudo-observations for realism despite exact constraint satisfaction.

Our contributions are: - A unified view of spatial motion control as a linear inverse problem solved by projection during flow sampling. - A kinematics-aware metric that respects skeletal topology, distributing corrections coherently to preserve realism. - A time-varying pseudo-observation schedule with dynamic masking and adaptive variance for robust zero-shot inpainting. - Exact constraint satisfaction without retraining or inner-loop optimization, with competitive realism to trained controllers.

Beyond motion, the same projection applies to any linear control over structured trajectories, suggesting utility for robotics path following and camera planning where exact feasibility is critical.

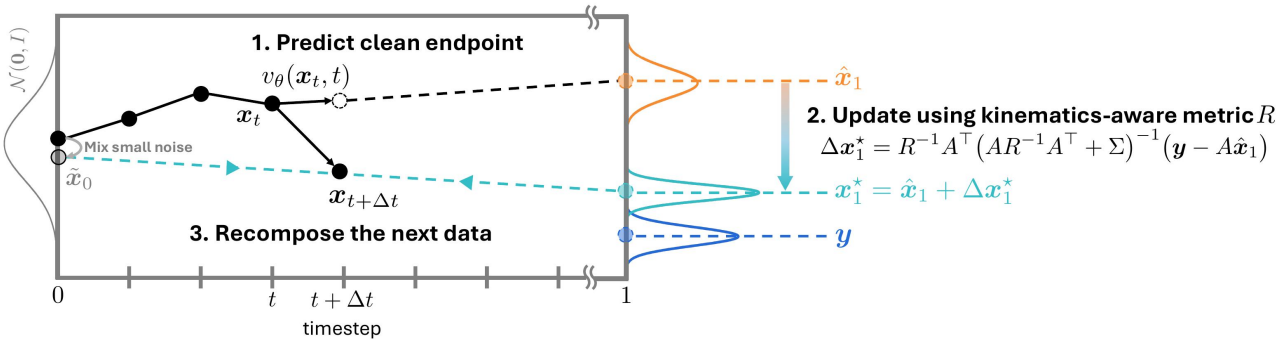

Figure 1. Overview of the projection sampling step. At each time, the clean endpoint estimated by the flow prior is projected onto the feasible set using a kinematics-aware metric, then stochastically recomposed to stay on-manifold.

2. Related Work

Motion generation. Diffusion-based human motion generation has achieved strong realism and flexibility[11, 12], with many extensions to conditioning and scene grounding. Flow-based formulations[1, 6, 7] offer an alternative view via ODEs and conditional probability paths, enabling efficient sampling with vector-field regression.

Spatial control. Pelvis-trajectory and multi-joint control have been tackled with guidance and controller networks, such as OmniControl[13] and Guided Motion Diffusion (GMD)[4]. Controller-style training (e.g., ControlNet[14]) and latent conditioning (e.g., ACMDM[8]) can improve realism but require retraining per task or signal. Inference-time noise optimization (DNO)[5] adapts a pre-trained prior without retraining but incurs iterative inner loops and still leaves residual errors.

Projection and inverse problems. Projection and proximal updates are established tools for solving linear inverse problems and imposing constraints within iterative solvers. Recently, inverse problem solvers with diffusion/flow priors have blended data-consistency with learned priors. ProjFlow brings a projection step into flow sampling for motion, leveraging a kinematics-aware metric to preserve naturalness while enforcing exact feasibility.

Beyond motion, projection-style data-consistency has been effective in many inverse problems with learned priors. Unlike guidance that perturbs the score or vector field, our update enforces feasibility exactly in the observation space and delegates realism to the prior, cleanly separating roles. This is especially natural for linear control signals such as trajectories, keypoints, and reprojections, where closed-form updates exist and integrate seamlessly into fast ODE samplers.

Discussion. Prior guidance-based controllers trade off feasibility and fidelity through continuous nudging of the score/vector field, which may leave small residual errors and entangle roles of data and prior. ProjFlow instead uses an explicit projection that satisfies measurements exactly and a kinematics-aware metric that propagates corrections along the skeleton, reducing foot sliding and jitter while avoiding retraining. This perspective complements controller training and noise optimization by offering a plug-in, training-free alternative.

3. Preliminaries

Flow-matching priors. Flow matching[1, 6] learns a time-dependent vector field that transports Gaussian noise to data along a chosen probability path (e.g., linear or displacement interpolants), enabling fast ODE-based sampling. Rectified flow[7] encourages straight transport paths. In our setup, the motion prior predicts at each time the clean endpoint estimate and residual noise consistent with a Tweedie-style decomposition, which we exploit for projection and stochastic recomposition.

Linear measurement model. Many spatial control objectives can be represented as linear measurements of the global joint trajectory. Let $\boldsymbol{x} \in \mathbb{R}^{3JT}$ denote the stacked global joint coordinates over T frames and J joints. A measurement operator $\boldsymbol{A}_t$ and observation $\boldsymbol{y}_t$ define a constraint $\boldsymbol{y}_t = \boldsymbol{A}_t\boldsymbol{x} + \boldsymbol{\epsilon}_t$, where $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_t)$ models noise and confidence. Examples include pelvis trajectory sampling, per-joint keyframe positions, and 2D reprojection with known camera.

4. Method

ProjFlow augments a flow-matching sampler with a closed-form projection step that enforces linear measurements at each time while preserving motion naturalness through a kinematics-aware metric and stochastic recomposition.

Step 1: Clean endpoint prediction. Given the current state and time t, the flow prior yields an estimate of the clean motion endpoint $\hat{\boldsymbol{x}}_t$ and an associated noise scale. We treat $\hat{\boldsymbol{x}}_t$ as the data-consistent target before projection.

Step 2: Kinematics-aware projection. We enforce

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measurements $\boldsymbol{y}_t = \boldsymbol{A}_t\boldsymbol{x}+\boldsymbol{\epsilon}_t$ by projecting $\hat{\boldsymbol{x}}_t$ onto the feasible set in a skeleton-aware metric. Specifically, we solve a regularized least-squares problem that trades off measurement fidelity and proximity to $\hat{\boldsymbol{x}}_t$ in an $\boldsymbol{R}$-norm: find $\tilde{\boldsymbol{x}}_t$ minimizing $\|\boldsymbol{A}_t\boldsymbol{x}-\boldsymbol{y}_t\|^2_{\Sigma_t^{-1}} + \|\boldsymbol{x}-\hat{\boldsymbol{x}}_t\|^2_{\boldsymbol{R}}$. The closed-form solution is

$$\tilde{\boldsymbol{x}}_t \;=\; \hat{\boldsymbol{x}}_t - \boldsymbol{R}^{-1}\boldsymbol{A}_t^\top\big(\boldsymbol{A}_t\boldsymbol{R}^{-1}\boldsymbol{A}_t^\top+\Sigma_t\big)^{-1}(\boldsymbol{A}_t\hat{\boldsymbol{x}}_t-\boldsymbol{y}_t). \tag{1}$$

Here $\boldsymbol{R}$ embeds skeletal topology via a Laplacian-based kinematics-aware metric so that corrections propagate coherently along the kinematic tree rather than creating local distortions. As $\Sigma_t \to \mathbf{0}$ the update enforces hard constraints exactly; finite Σ_t handles noisy measurements.

Step 3: Stochastic recomposition. To remain on the learned manifold and avoid overfitting to constraints, we mix the corrected endpoint with noise according to the time-dependent prior noise schedule, yielding the next sample state consistent with the flow dynamics.

Pseudo-observations for sparse inputs. For sparse or long-gap constraints (e.g., inpainting), a fixed $\boldsymbol{A}_t$ may be too weak early in sampling. We therefore introduce a time-varying operator with dynamic masking and adaptive variance: early times include randomly masked pseudo-observations with larger Σ_t to gently steer sampling, tightening masks and variances as t decreases. This prevents ignoring sparse signals and improves stability.

Supported tasks. ProjFlow supports trajectory control, multi-joint keyframe matching, 2D-to-3D lifting with known cameras (by letting $\boldsymbol{A}_t$ be the projection operator), and inpainting via time-scheduled masks—all without retraining or inner-loop optimization.

Complexity and stability. Each projection solves a $K\times K$ system over measured dimensions with $\boldsymbol{A}_t\boldsymbol{R}^{-1}\boldsymbol{A}_t^\top+\Sigma_t$. In practice K is moderate, $\boldsymbol{A}_t$ is sparse, and Σ_t is diagonal, so a Cholesky or preconditioned CG adds negligible overhead relative to the base step. We precompute sparse factorizations for the Laplacian-based $\boldsymbol{R}^{-1}$ and cache per-task $\boldsymbol{A}_t$ patterns. The time-varying Σ_t schedule smooths early updates, avoids oscillations, and preserves exact feasibility as $t\to 0$.

Implementation details. We construct $\boldsymbol{R}$ from a normalized kinematic Laplacian with bone-length weighting and add a small diagonal for numerical stability; per-joint rescaling balances root and limb influence. The variance schedule Σ_t anneals from a large value at early times to near-zero at late times with a cosine cap to avoid overshooting; the same schedule is shared across tasks unless noted. Operators $\boldsymbol{A}_t$ are sparse and assembled on-the-fly from task descriptors (trajectories, keyframes, camera matrices), and we solve the $K\times K$ systems in batch using cached sparsity patterns.

Connection to proximal sampling. Our update is the proximal map of the measurement loss under the kinematic metric. Under linear–Gaussian assumptions it equals the posterior mean, yielding a statistically optimal, closed-form data-consistency step and a principled way to anneal Σ_t within flow matching.

5. Experiments

Setup. We evaluate on HumanML3D[3] using an ACMDM flow prior[8] as the base motion model. We report realism (FID), text-motion metrics (R-precision, Matching distance), and control accuracy (Traj./Loc./Avg. errors, MPJPE-2D/3D) following common practice and prior work[4, 13, 14].

Evaluation protocol. We follow the official HumanML3D splits and text prompts, generate 32 samples per prompt unless specified, and compute FID using the standard motion feature extractor with bootstrap over seeds. Control errors are measured in global coordinates under dataset conventions; for 2D-to-3D we use provided camera intrinsics and check exact reprojection consistency. All methods use the same base prior and sampling steps for fairness.

Baselines and hyperparameters. We reuse official checkpoints and their conditioning interfaces for trained controllers; for DNO-style optimization we tune step counts and guidance weights on validation prompts and early-stop by control error. All methods use identical 80-step schedules; ProjFlow adds one sparse $K\times K$ solve per step. **Sampling budget and seeds.** Unless otherwise stated we use 80 ODE steps and fix random seeds across methods to reduce variance during FID computation; increasing steps to 100–120 yields only marginal realism changes while control error remains exactly zero by construction. We release scripts to reproduce all metrics with the exact evaluation splits.

Method overview figure. An overview of the projection sampling step is shown in figure 1.

Trajectory and multi-joint control. Table 1 compares ProjFlow with trained controllers and optimization-based baselines across pelvis and all-joint settings. ProjFlow achieves exact constraint satisfaction (0.0000 Traj./Loc./Avg.) while maintaining low FID. For pelvis control, ProjFlow reaches FID 0.107 and zero control error, outperforming DNO in FID and eliminating residual errors.

2D-to-3D lifting with text conditioning. ProjFlow enforces exact 2D reprojection (MPJPE-2D $=0.000$) while preserving realism. Quantitative comparisons under Average/Cross settings are summarized in Table 2.

Qualitative results. Figure 2 illustrates text-conditioned pelvis-trajectory control; Figure 3 shows long-gap inpainting guided by pseudo-observations; Figure 4 depicts 2D-to-3D hand-trajectory lifting.

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Table 1. **Quantitative text-conditioned motion generation with spatial control signals and upper-body editing on HumanML3D[3].** In the first section, methods are trained and evaluated solely on pelvis controls. In the middle section, methods are trained on all joints and evaluated separately on each controlled joint. Only average results are reported for brevity. We include details in the supplementary material. The last section presents upper-body editing results. **bold** face / underline indicates the best/2nd results.

Controlling Joint	Methods	Zero-shot?	FID↓	R-Precision Top 3	Diversity→	Foot Skating Ratio.↓	Traj. err.↓	Loc. err.↓	Avg. err.↓
	GT	-	0.000	0.795	10.455	-	0.000	0.000	0.000
Pelvis	MDM [12]	✓	1.792	0.673	9.131	0.1019	0.4022	0.3076	0.5959
	PriorMDM [10]	✗	0.393	0.707	9.847	0.0897	0.3457	0.2132	0.4417
	GMD [4]	✓	0.238	0.763	10.011	0.1009	0.0931	0.0321	0.1439
	OmniControl [13]	✗	0.081	0.789	10.323	0.0547	0.0387	0.0096	0.0338
	MotionLCM V2+CtrlNet [2]	✗	3.978	0.738	9.249	0.0901	0.1080	0.0581	0.1386
	MaskControl [9]	✗	**0.066**	0.799	**10.474**	**0.0543**	**0.0000**	**0.0000**	0.0093
	ACMDM-S-PS22+CtrlNet [8]	✗	0.067	**0.805**	10.481	0.0591	0.0075	0.0010	0.0100
	ACMDM-S-PS22+DNO [5]	✓	0.151	0.802	–	0.0610	0.0027	0.0002	0.0089
	ACMDM-S-PS22+ProjFlow	✓	0.107	0.784	10.644	0.0629	**0.0000**	**0.0000**	**0.0000**
All Joints (Average)	OmniControl [13]	✗	0.126	0.792	10.276	0.0608	0.0617	0.0107	0.0404
	MotionLCM V2+CtrlNet [2]	✗	4.504	0.715	9.230	0.1119	0.2740	0.1315	0.2464
	MaskControl [9]	✗	0.095	0.795	10.159	**0.0545**	**0.0000**	**0.0000**	0.0065
	ACMDM-S-PS22+CtrlNet [8]	✗	**0.070**	**0.803**	**10.526**	0.0596	0.0117	0.0019	0.0197
	ACMDM-S-PS22+DNO [5]	✓	0.147	0.800	–	0.0600	0.0034	0.0003	0.0121
	ACMDM-S-PS22+ProjFlow	✓	0.097	0.779	10.651	0.0603	**0.0000**	**0.0000**	**0.0000**

	Methods	Zero-shot?	FID↓	R-Precision Top 1	R-Precision Top 2	R-Precision Top 3	Matching↓	Diversity→	–
Upper-Body Edit	MDM [12]	✓	1.918	0.359	0.556	0.654	4.793	9.210	–
	OmniControl [13]	✗	0.909	0.428	0.614	0.722	3.694	10.207	–
	MotionLCM V2+CtrlNet [2]	✗	3.922	0.404	0.592	0.692	5.610	9.309	–
	MaskControl [9]	✗	**0.066**	0.501	0.695	0.794	3.227	10.159	–
	ACMDM-S-PS22+CtrlNet [8]	✗	0.076	**0.532**	**0.719**	**0.820**	**3.098**	10.586	–
	ACMDM-S-PS22+ProjFlow	✓	0.087	0.501	0.690	0.787	3.319	**10.571**	–

Table 2. **Quantitative analysis of ProjFlow and three baseline models proposed in Sketch2Anim [15] on the HumanML3D [3].** Evaluation metrics on motion realism, control accuracy, and text-motion match are presented. Following OmniControl [13], we report both the average error of all joints (Average) and their random combination (Cross). **bold** face / underline indicates the best/2nd results.

Condition	Method	Realism		Control Accuracy				Text-Motion Matching	
		FID ↓	Foot Skating ↓	MPJPE-2D ↓	MPJPE-3D ↓	Avg. Err.-2D ↓	Avg. Err.-3D ↓	Matching↓	R-precision (Top-3) ↑
Average	Motion Retrieval	0.690	**0.064**	0.057	0.076	0.290	0.410	4.060	0.640
	Lift-and-Control	0.979	0.089	0.054	0.071	0.261	0.340	3.297	0.752
	Direct 2D-to-Motion	2.553	0.112	0.040	0.055	0.193	0.275	3.723	0.687
	Sketch2Anim [15]	0.525	0.103	0.036	0.048	0.087	**0.134**	**3.077**	**0.802**
	ACMDM-S-PS22+ProjFlow	**0.349**	0.146	**0.000**	**0.042**	**0.000**	0.331	3.363	0.748
Cross	Motion Retrieval	**0.103**	**0.067**	0.055	0.073	0.307	0.423	3.405	0.724
	Lift-and-Control	0.738	0.101	0.051	0.067	0.209	0.283	3.135	0.778
	Direct 2D-to-Motion	2.310	0.123	0.040	0.056	0.189	0.266	3.606	0.709
	Sketch2Anim [15]	0.577	0.102	0.033	0.046	0.079	**0.132**	**3.042**	**0.796**
	ACMDM-S-PS22+ProjFlow	0.168	0.139	**0.000**	**0.037**	**0.000**	0.298	3.259	0.764

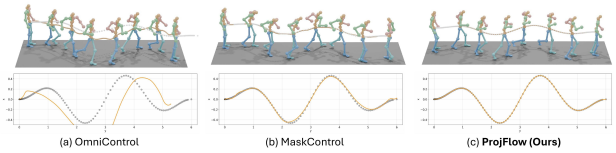

Figure 2. Text-conditioned pelvis-trajectory control.

6. Analysis

Robustness and cost. Across seeds control error stays 0.0000 with little FID variation, indicating recomposition preserves feasibility. Overhead is modest since projection solves sparse systems with cached factorizations; runtime scales mainly with the number of enforced measurements.

Ablations. Table 3 quantifies the roles of our components. Replacing the kinematics-aware metric with Euclidean ($\boldsymbol{R}$=I) worsens FID to 1.152; removing stochastic recomposition increases FID to 3.429; and removing pseudo-observations yields FID 0.880. All ablations maintain 0.0000 control error, indicating that realism gains derive from topology-aware corrections, on-manifold noise mixing, and time-varying guidance.

Per-setting behavior. In the Average vs. Cross settings of 2D-to-3D lifting (Table 2), ProjFlow preserves exact 2D consistency while achieving strong FID and R-precision, demonstrating robust generalization across joint subsets. **Schedule and metric sensitivity.** Results are stable under cosine or linear variance annealing; too slow decay can un-

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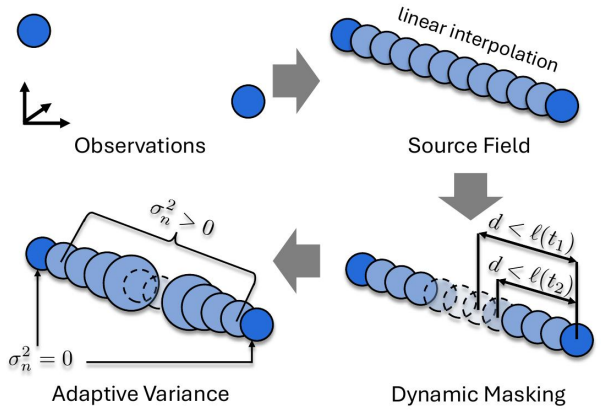

Figure 3. Pseudo-observations for motion inpainting.

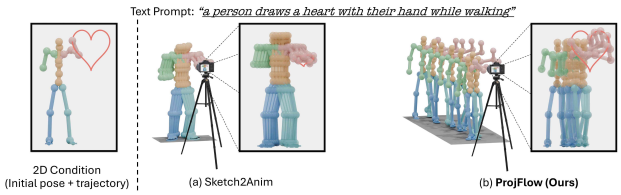

Figure 4. 2D-to-3D hand-trajectory lifting with text conditioning.

Table 3. Ablation studies of ProjFlow.

Variant	FID↓	R-Prec.	Div.→	Foot↓	Traj.↓	Loc.↓	Avg.↓
ProjFlow (Full)	0.097	0.779	10.651	0.0603	0.0000	0.0000	0.0000
Euclid. (R=I)	1.152	0.740	10.107	0.0595	0.0000	0.0000	0.0000
No noise (η_t=0)	3.429	0.707	9.307	0.0863	0.0000	0.0000	0.0000
Plain masking	0.880	0.748	10.187	0.0632	0.0000	0.0000	0.0000

derweight measurements early on, while overly aggressive late-time variance may induce small oscillations despite exact feasibility. Replacing the Laplacian kinematic metric with a diagonal bone-weighted metric preserves feasibility but slightly increases foot sliding and temporal jitter in long sequences.

7. Conclusion

We presented ProjFlow, a projection-based sampler for flow-matching motion priors that achieves zero-shot exact spatial control by casting control as a linear inverse problem and enforcing feasibility via kinematics-aware projections with stochastic recomposition. Across trajectory control, multi-joint editing, and 2D-to-3D lifting on HumanML3D, ProjFlow attains 0.0000 control error with competitive or superior realism compared to optimization-based and trained baselines, all without retraining or inner-loop optimization. The results suggest projection-driven sampling is an effective paradigm for precise, interactive motion authoring.

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